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## **EXNO 1: Estimation of Characteristic Impedance ( $Z_0$ ), and Propagation Constant ( $\gamma$ ) of a Transmission Line from Primary Constants**

### **Test Case 1:**

Investigating the significance of Characteristic Impedance ( $Z_0$ ), and Propagation Constant ( $\gamma$ ) of a Transmission Line using the following parameters

- a) Resistance =  $10 \Omega/\text{Km}$
- b) Capacitance =  $8 \mu\text{F}/\text{Km}$
- c) Inductance =  $3.5 \text{ mH}/\text{Km}$
- d) Conductance =  $4 \mu\text{mho}/\text{Km}$
- e) Frequency =  $2 \text{ KHz}$

**AIM:** Investigating the significance of Characteristic Impedance ( $Z_0$ ), and Propagation Constant ( $\gamma$ ) of a Transmission Line with Resistance =  $10 \Omega/\text{Km}$ , Capacitance =  $8 \mu\text{F}/\text{Km}$ , Inductance =  $3.5 \text{ mH}/\text{Km}$ , Conductance =  $4 \mu\text{mho}/\text{Km}$ , Frequency =  $2 \text{ KHz}$

### **SOFTWARE REQUIRED:**

SCILAB Version 6.1.1

### **PROCEDURE:**

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

### **PROGRAM:**

```

clear; clc;

R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);

w=2*pi*f;

Z=R+(%i*w*L);

Y=G+(%i*w*C);

Zo=sqrt(Z/Y);

C=round(real(Zo));

D=round(imag(Zo));

printf('-Zo = %f + j(%f)ohms\n',C,D);

P=sqrt(Z*Y);

a=real(P);

a1=round(a*10000)/10000;

printf('-Attenuation constant a = %f neper/km\n',a1);

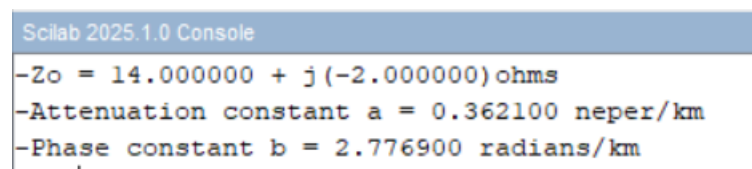
b=imag(P);

b1=round(b*10000)/10000;

printf('-Phase constant b = %f radians/km',b1);

```

### OUTPUT:



```

Scilab 2025.1.0 Console
-Zo = 14.000000 + j(-2.000000)ohms
-Attenuation constant a = 0.362100 neper/km
-Phase constant b = 2.776900 radians/km

```

### RESULT:

The characteristic impedance and propagation constant of a transmission line is calculated using SCILAB and the impact of primary constants on secondary constants are investigated.

## Test Case 2:

Examine the impact of resistance and inductance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line at a frequency  $f=2.5$  KHz. Vary the resistance value from  $10\ \Omega$  to  $50\ \Omega$  at the interval of  $10\ \Omega$  with L,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the inductance values from  $3.5\text{ mH}$  to  $11.5\text{ mH}$  at the interval of  $2.5\text{ mH}$  with R,G and C values constant.

**AIM:** To examine the impact of resistance and inductance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line at a frequency  $f=2.5$  KHz. Vary the resistance value from  $10\ \Omega$  to  $50\ \Omega$  at the interval of  $10\ \Omega$  with L,G and C values constant and calculate the Characteristic Impedance and Propagation Constant. Vary the inductance values from  $3.5\text{ mH}$  to  $11.5\text{ mH}$  at the interval of  $2.5\text{ mH}$  with R,G and C values constant.

## SOFTWARE REQUIRED:

SCILAB Version 6.1.1

## PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

## PROGRAM:

```
clear; clc;  
R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);  
w=2*%pi*f;  
Z=R+(%i*w*L);  
Y=G+(%i*w*C);
```

```

Zo=sqrt(Z/Y);
C=round(real(Zo));
D=round(imag(Zo));
printf('-Zo = %f + j(%f)ohms\n',C,D);
P=sqrt(Z*Y);
a=real(P);
a1=round(a*10000)/10000;
printf('-Attenuation constant a = %f neper/km\n',a1);
b=imag(P);
b1=round(b*10000)/10000;
printf('-Phase constant b = %f radians/km',b1);

```

### OUTPUT:

Variable R

Constants L=3mH; f=2000; G=.4\*(10<sup>-6</sup>); C=8\*(10<sup>-6</sup>).

| S.No. | Inputs(R) | Z0(ohms)                  | $\alpha$ (neper/km) | $\beta$ (radians/km) |
|-------|-----------|---------------------------|---------------------|----------------------|
| 1     | 10        | 14.000000 + j(-2.000000)  | 0.362100            | 2.776900             |
| 2     | 30        | 15.000000 + j(-5.000000)  | 1.026500            | 2.938300             |
| 3     | 50        | 16.000000 + j(-8.000000)  | 1.582800            | 3.175700             |
| 4     | 70        | 17.000000 + j(-10.000000) | 2.050100            | 3.432600             |
| 5     | 90        | 18.000000 + j(-12.000000) | 2.453500            | 3.687700             |

Variable L

Constants R=10 $\Omega$ ; f=2000; G=.4\*(10<sup>-6</sup>); C=8\*(10<sup>-6</sup>).

| S.No. | Inputs(L)          | Z0(ohms)                 | $\alpha$ (neper/km) | $\beta$ (radians/km) |
|-------|--------------------|--------------------------|---------------------|----------------------|
| 1     | 3*10 <sup>-3</sup> | 14.000000 + j(-2.000000) | 0.362100            | 2.776900             |
| 2     | 4*10 <sup>-3</sup> | 16.000000 + j(-2.000000) | 0.314700            | 3.194600             |
| 3     | 5*10 <sup>-3</sup> | 18.000000 + j(-1.000000) | 0.282000            | 3.565500             |
| 4     | 6*10 <sup>-3</sup> | 19.000000 + j(-1.000000) | 0.257700            | 3.902100             |
| 5     | 7*10 <sup>-3</sup> | 21.000000 + j(-1.000000) | 0.238700            | 4.212300             |

### RESULT:

The impact of resistance and inductance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line is examined at a frequency f=2.5 KHz.

### Test Case 3:

Explore the impact of frequency and capacitance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant . Vary the capacitance values from 8  $\mu$ F to 16  $\mu$ F at the interval of 2  $\mu$ F with R,G and L values constant.

**AIM:** To explore the impact of frequency and capacitance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line. Vary the frequency from 2KHz to 10 KHz at the interval of 2 KHz with L,R ,G and C values constant and calculate the Characteristic Impedance and Propagation Constant

### SOFTWARE REQUIRED:

SCILAB Version 6.1.1

### PROCEDURE:

- Open SCILAB software
- Open file-new
- Put the input values
- Type the program
- Execute the program
- End the program

### PROGRAM:

```
clear; clc;

R=10;L=0.003;f=2*10^3;G=4*(10^-6);C=16*(10^-6);

w=2*pi*f;

Z=R+(%i*w*L);

Y=G+(%i*w*C);

Zo=sqrt(Z/Y);

C=round(real(Zo));

D=round(imag(Zo));
```

```

printf('-Zo = %f + j(%f)ohms\n',C,D);

P=sqrt(Z*Y);

a=real(P);

a1=round(a*10000)/10000;

printf('-Attenuation constant a = %f neper/km\n',a1);

b=imag(P);

b1=round(b*10000)/10000;

printf('-Phase constant b = %f radians/km',b1);

```

## OUTPUT:

Variable f

Constants L=3mH; R=10; G=.4\*(10<sup>-6</sup>); C=16\*(10<sup>-6</sup>).

| S.No. | Inputs(f)<br>Hz | Z0(ohms)                 | $\alpha$ (neper/km) | $\beta$ (radians/km) |
|-------|-----------------|--------------------------|---------------------|----------------------|
| 1     | 1000            | 14.000000 + j(-4.000000) | 0.353700            | 1.421300             |
| 2     | 2000            | 14.000000 + j(-2.000000) | 0.362100            | 2.776900             |
| 3     | 3000            | 14.000000 + j(-1.000000) | 0.363800            | 4.145700             |
| 4     | 4000            | 14.000000 + j(-1.000000) | 0.364400            | 5.518300             |
| 5     | 5000            | 14.000000 + j(-1.000000) | 0.364700            | 6.892500             |

Variable C

Constants R=10; L=.009; f=2500; G=.4\*(10<sup>-6</sup>);

| S.No. | Inputs(C) F         | Z0(ohms)                 | $\alpha$ (neper/km) | $\beta$ (radians/km) |
|-------|---------------------|--------------------------|---------------------|----------------------|
| 1     | 16*10 <sup>-6</sup> | 24.000000 + j(-1.000000) | 0.210700            | 5.964500             |
| 2     | 17*10 <sup>-6</sup> | 23.000000 + j(-1.000000) | 0.217200            | 6.148000             |
| 3     | 18*10 <sup>-6</sup> | 22.000000 + j(-1.000000) | 0.223500            | 6.326300             |
| 4     | 19*10 <sup>-6</sup> | 22.000000 + j(-1.000000) | 0.229600            | 6.499600             |
| 5     | 20*10 <sup>-6</sup> | 21.000000 + j(-1.000000) | 0.235600            | 6.668500             |

## RESULT:

The impact of frequency and capacitance on the significance of Characteristic Impedance ( $Z_0$ ) and Propagation Constant ( $\gamma$ ) of a Transmission Line is investigated for various values of frequencies and capacitance values.